

AMIR JOUDAKI

Fundamental AI/ML researcher

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PRACTICAL EXPERIENCE

Research Scientist

ETH Zurich

Nov 2024 – now Zürich, Switzerland

I work towards a comprehensive understanding of how deep nonlinear models learn, and more recently on barriers to learning in real world tasks with task and distributional changes, as well as reasoning. I also supervise a number MSc. students.

Lead Research Collaborator

ETH Zurich & Apple Inc.

Nov 2024 – now Zürich, Switzerland

Contributed to a fundamental research on barriers for learning in real world applications that distributions shift. Developed a novel mathematical framework to predict how and why deep neural networks lose their ability to adapt. Design and implementation of comprehensive experiments on wide-ranging tasks and architectures.

Ph.D. in Artificial Intelligence

ETH Zurich

Sept 2019 – Nov 2024 Zürich, Switzerland

Authored 8 papers (6 first authored) in top-tier venues in AI/ML and computational genomics. Supervised >10 MSc thesis/semester projects, leading to several top-tier publications, organized teaching.

Machine Learning Consultant

SkillLab

Jan 2022 – July 2022 Zürich, Switzerland

Research and development of a proof-of-concept for training, labeling, and monitoring NLP tasks, written in PyTorch, Flask, Typescript, React, MongoDB, orchestrated as micro-services in REST API, and run inside docker containers for deployment.

EDUCATION

Direct Ph.D. in Artificial Intelligence

ETH Zurich

Feb 2017 – Nov 2024 Zürich, Switzerland

Direct PhD is a highly selective program (<5% of all PhD admissions), with 2 years of MSc courses followed by 4 years of PhD research.
Ph.D. supervisors: Gunnar Ratsch & Francis Bach.
MSc Thesis: Scalable algorithms for biological sequence analysis

M.Sc. in Cognitive Neuroscience

International School for Advanced Studies (SISSA)

Feb 2014 – Jan 2017 Trieste, Italy

Thesis: Modeling activity of electrophysiological recordings in vivo in rats

B.Sc. in Computer Engineering

Sharif University of Technology

Sept 2008 – Sept 2011 Tehran, Iran

PROGRAMMING

PythonPyTorchJAX

Scikit-LearnNumpy

C/C++OpenMPMPI

CUDANumbaDocker

SparkSQLBashGit

JavaLaTeX

AWARDS

Direct Doctorate Fellowship
Among two out of more than one hundred students selected for direct PhD program at ETH computer science department, 2017

Ranked 42nd
Among more than 50,000 participants in National Higher Education Entrance Exam, 2011.

Ranked 369th
Among more than 400,000 participants in National University Entrance Exam, 2007.

Best Reviewer Award
Best reviewer AISTATS 2025, Top reviewer ICML 2025

LANGUAGES

English: Proficient
Persian: Native
German: Beginner

REFERENCES

Thomas Hofmann (ETH)
Francis Bach (ENS)
Gunnar Ratsch (ETH)
Mehrdad Farajtabar (Apple)

PUBLICATIONS

Mathematical foundations of AI

- [Amir Joudaki](#), Giulia Lanzillotta, Mohammad Samragh Razlighi, Iman Mirzadeh, Keivan Alizadeh, Thomas Hofmann, Mehrdad Farajtabar, Fartash Faghri, "Barriers for Learning in an Evolving World: a Mathematical Understanding of Loss of Plasticity", (submitted, under review)
- [Amir Joudaki](#), Thomas Hofmann, "Emergence of globally attracting fixed points in deep neural networks with nonlinear activations", AISTATS (poster).
- Alex Meterez*, [Amir Joudaki](#)*, Francesco Orabona, Alex Immer, Gunnar Ratsch, Hadi Daneshmand, "Batch normalization without gradient explosion: Towards training without depth limits", accepted to proceedings of ICLR 2024 (poster). (*: equally contributed).
- [Amir Joudaki](#), Hadi Daneshmand Francis Bach, "On the impact of activation and normalization in obtaining isometric embeddings at initialization", NeurIPS 2023 (poster).
- [Amir Joudaki](#), Hadi Daneshmand Francis Bach, "On Bridging the Gap between Mean Field and Finite Width in Deep Random Neural Networks with Batch Normalization", ICML 2023 (poster).
- Hadi Daneshmand, [Amir Joudaki](#), Francis Bach, "Batch Normalization Orthogonalizes Representations in Deep Random Networks", **spotlighted NeurIPS 2021 (top 3% of submissions)**.
- Alexandre Bense, Amir Joudaki, Tim G. J. Rudner, Vincent Fortuin, "PCA Subspaces Are Not Always Optimal for Bayesian Learning ", NeurIPS 2021 workshop (DistShift).

Genomics Sequence Analysis

- Flavia Pedrocchi, Stefan Stark, Gunnar Ratsch, [Amir Joudaki](#), "Identifying Biological Priors and Structure in Single-Cell Foundation Models", ICML workshop 2024 (Efficient and Accessible Foundation Models for Biological Discovery).
- Kacper Kapusniak, Manuel Burger, Gunnar Ratsch, [Amir Joudaki](#), "Learning Genomic Sequence Representations using Graph Neural Networks over De Bruijn Graphs", NeurIPS workshop 2023 (Graph Learning Frontiers).
- [Amir Joudaki](#)*, Alexandru Meterez*, Harun Mustafa, Ragnar Groot Koerkamp, Andre Kahles, Gunnar Raetsch, "Aligning distant sequences to graphs using long seed sketches", Journal of Genome Research 2023 (Cold Spring Harbor Laboratory press). (*: equally contributed).
- [Amir Joudaki](#), Gunnar Ratsch, Andre Kahles, "Fast Alignment-Free Similarity Estimation By Tensor Sketching", proceedings of RECOMB 2021.
- Mikhail Karasikov, Harun Mustafa, [Amir Joudaki](#), Sara Javadzadeh-No, Gunnar Rätsch, André Kahles, "Sparse Binary Relation Representations for Genome Graph", Journal of computational biology: a journal of computational molecular cell biology 27.4 (2020): 626-639.

Dimensionality Reduction

- Amir Najafi, [Amir Joudaki](#), "Nonlinear Dimensionality Reduction via Path-Based Isometric Mapping", IEEE transactions on pattern analysis and machine intelligence, 38(7), 1452-1464.

Functional Brain Networks Analysis

- [Amir Joudaki](#)*, Elham Barzegaran*, Mahdi Jalili*, Andrea O. Rossetti, Richard S. Frackowiak, and Maria G. Knyazeva, "Properties of functional brain networks affect frequency of psychogenic non-epileptic seizures", Frontiers in Human Neuroscience, 6:335. (*: equally contributed).
- [Amir Joudaki](#), Niloufar Salehi, Mahdi Jalili, Maria Knyazeva, "EEG-based functional brain networks: does the network size matter?", PLoS ONE 7(4): e35673.

SUPERVISION/MENTORSHIP

Current students:

- Guiv Farmanfarmayan, working on "Improving the efficiency of RL-based reasoning"
- Annalisa Belloni, working on "Improving cloning models with insights from loss of plasticity"
- Flavia Pedrocchi, working on "Interpreting single cell foundation models using sparse auto-encoders"

Former students and mentorships:

- Alex Meterz, PhD student at Harvard university
- Alec Flowers, Nvidia
- Kacper Kapusniak, PhD student at Harvard